

Clinically Reliable Privacy-Preserving Federated Learning under Heterogeneous ICU Environments

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Abstract—Purpose: Privacy regulations (HIPAA, GDPR) prevent hospitals from pooling patient records for joint model training, yet isolated local datasets limit predictive model quality. This paper evaluates whether federated learning (FL) can deliver clinically reliable ICU mortality prediction without centralizing sensitive records, and whether Byzantine-robust aggregation can protect model quality when hospital nodes are compromised.

Methods: We partition MIMIC-IV (65,273 admissions, 31 clinical features) across 7 ICU care units to simulate a heterogeneous multi-client network with realistic Non-IID populations. Five aggregation strategies are evaluated: FedAvg, FedProx, Median, Krum, and F_2 -score-weighted aggregation (FedF2). Client-side differential privacy is enforced via per-sample DP-SGD ($\epsilon=4.36$, $\delta=10^{-5}$, verified via moments accountant). Calibrated decision thresholds are optimized for Recall given asymmetric clinical costs. External validation is performed on an independent eICU-CRD cohort (22,361 patients across 7 hospitals).

Results: The centralized baseline achieves 85.2% Recall (AUROC 0.8920). Federated XGBoost via soft voting attains AUROC 0.9155 (<1% loss), while federated logistic regression reaches AUROC 0.8784 after Platt calibration. Byzantine-robust aggregation (Median, Krum) maintains AUROC ≥ 0.8594 under severe 3-of-7 client poisoning, whereas standard FedAvg collapses to AUROC 0.5011 (43% loss). DP-SGD yields AUROC 0.8477 post-calibration. External validation on eICU-CRD confirms generalization (AUROC 0.8337, 1.23% federated loss).

Conclusion: Privacy-preserving federated learning with Byzantine-robust aggregation is viable for multi-hospital critical care networks. Byzantine-robust methods (Median, Krum) are essential when client integrity cannot be guaranteed, whereas FedAvg suffices only in fully trusted settings. These results offer a reproducible benchmark for clinical FL deployment.

Index Terms—Federated Learning, Healthcare Machine Learning, Differential Privacy, Non-IID Data, Byzantine Robustness

I. INTRODUCTION

Clinical facilities today generate patient data at an unprecedented rate—electronic records, imaging archives, laboratory panels—but almost none of it flows between institutions [1]. The reasons are well known. Data protection frameworks like the Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in the European Union explicitly forbid shipping raw patient records to any external server [2], [3]. At the same time, ransomware campaigns targeting hospital

IT infrastructure have multiplied, making even authorized centralized storage increasingly risky [4]. The net effect is that every clinic holds its own data island. Machine learning models that need broad, multi-site training sets to generalize simply cannot access them.

Federated Learning (FL) was proposed precisely to work around this constraint [5]. The idea is straightforward: data never moves. Instead, every participating hospital trains a local copy of the model and transmits only the resulting weight updates to a central server, which fuses them into a shared global model [6]. No individual patient record ever leaves its home institution, so the scheme is compatible with HIPAA and GDPR by construction. Researchers have already tested variants of this approach on electronic health records [7], wearable physiological monitoring [1], and multi-hospital clinical trials [8], [9].

Moving from controlled experiments to actual hospital networks, however, uncovers a stubborn problem. Most FL algorithms—Federated Averaging (FedAvg) chief among them—were designed with well-curated, Independent and Identically Distributed (IID) benchmarks in mind [10]. Real clinical populations look nothing like that. A diabetes clinic in a metropolitan area and a rural general practice serve wildly different patient demographics; their local datasets are, statistically speaking, deeply Non-Independent and Identically Distributed (Non-IID) [11]. When a server naively averages weights from such mismatched clients, the optimization process can destabilize: different hospitals effectively tug the model in opposite directions [12], [13]. In healthcare, where disease class ratios vary enormously from site to site, this problem is especially acute [14].

There is also a security angle that deserves attention. FL does avoid moving raw data, but the gradient updates that *do* travel across the network are not harmless metadata. Zhu et al. demonstrated that, under favorable conditions, an adversary can mathematically invert shared gradients to reconstruct the original training records [15]. Unless the system enforces mathematically grounded privacy limits, this vulnerability persists [16], [17].

Motivated by these overlapping challenges, we built and evaluated a privacy-hardened FL pipeline on MIMIC-IV, a large-scale intensive care database. Within MIMIC-IV, we partition data by ICU care unit to simulate heterogeneous client populations, and validate generalization externally on the eICU-CRD dataset spanning 7 geographically independent hospitals. Our three main contributions are:

- We simulate realistic client heterogeneity through ICU care-unit partitioning within MIMIC-IV, which produces

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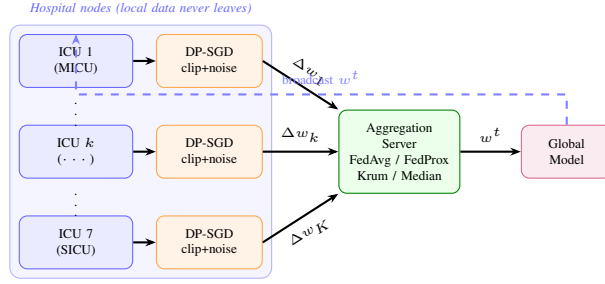


Fig. 1. Federated clinical learning architecture: 7 ICU care units train local mortality prediction models with DP-SGD (per-sample gradient clipping + calibrated Gaussian noise). Privatized model updates Δw_k are sent to a central aggregation server, which applies Byzantine-robust aggregation (FedAvg, FedProx, Krum, or Median) and broadcasts the global model w^t back to clients. No raw patient data or EHR records are transmitted. Aggregation server can tolerate compromised clients (e.g., due to ransomware or misconfiguration) by using robust aggregation methods.

Non-IID populations comparable to those seen across hospital networks. We explicitly acknowledge that all 7 clients originate from a single institution (Beth Israel Deaconess Medical Center); the eICU-CRD external validation (Section IV-I) provides genuine multi-hospital evaluation across 7 independent hospitals.

- We conduct the first comprehensive empirical evaluation of Byzantine-robust aggregation (Median, Krum) under multiple attack types (label-flip, sign-flip, adaptive) and attacker counts (1–3 of 7 clients) in a clinical federated learning setting. Key finding: Median and Krum maintain $\text{AUROC} \geq 0.8594$ even under severe 3-client poisoning (Table IV), while standard FedAvg degrades catastrophically to $\text{AUROC} 0.5011$ (43% loss). We also document that FedProx collapses under label-flip attacks ($\text{AUROC} 0.2953$), a result we attribute to its proximal regularization preventing recovery from poisoned initializations and which should be reproduced with μ ablation before deployment.
- We validate privacy–utility tradeoffs (DP-SGD at $\epsilon = 4.36$, $\delta = 10^{-5}$ achieves $\text{AUROC} 0.8477$ post-calibration), multi-architecture federated learning (XGBoost maintains 99.0% centralized AUROC via federated soft voting), and calibration requirements (Platt scaling recovers ECE from 0.23 to < 0.01). We also provide empirical guidance on F_2 -score-weighted aggregation (FedF2), an instantiation of existing performance-weighted FL, confirming it offers marginal calibration benefits in trusted networks but is insufficient under Byzantine attacks.

The rest of the paper is organized as follows. Section II reviews the relevant literature. Section III describes our experimental setup—data partitioning, aggregation rules, and privacy mechanisms. Section IV presents the results, including scalability experiments. Section V concludes the paper and suggests future directions.

II. RELATED WORK

The appeal of FL for healthcare is obvious: hospitals can collaborate on model training without ever handing over their patient data. Several groups have shown this works in practice. Rieke et al. [18] laid out a roadmap for FL in digital medicine; Sheller et al. [19] demonstrated federated

brain tumour segmentation where no imaging data crossed institutional lines; and more recently, Biswas et al. [20] paired FL with explainable AI to screen for pneumonia on GAN-augmented chest X-rays. Large-scale deployments have validated the clinical viability of the approach: Dayan et al. [21] coordinated federated COVID-19 outcome prediction across 20 hospitals, demonstrating that multi-institutional FL is operationally achievable in practice; Pati et al. [22] showed federated rare cancer segmentation across 71 sites; and researchers have applied FL to stroke outcome prediction [23], drug discovery [23], and electronic health records [7]. Broad surveys by Nguyen et al. [1] and Antunes et al. [8] paint a similarly encouraging picture, noting clear gains in multi-site data utilization and regulatory alignment with healthcare privacy frameworks. But both surveys also flag an uncomfortable truth: off-the-shelf optimization methods tend to break down once clinical data enters the picture. Xu et al. [7] saw this firsthand—their federated EHR pipeline worked well under controlled conditions but lost predictive power as soon as the participating hospitals differed significantly in patient demographics.

Class imbalance is the primary source of optimization instability in federated healthcare settings. One hospital might see mostly healthy patients; another might specialize in the very condition the model is trying to detect. The resulting local datasets are emphatically Non-IID. Under these conditions, the convergence guarantees of FedAvg no longer hold [24]. Zhao et al. [11] measured the damage in a controlled experiment: accuracy dropped by as much as 55% compared to a centralized IID baseline. The field has proposed fixes—FedProx adds a regularization penalty to prevent local models from wandering too far from the global state [12], while SCAFFOLD uses control variates to compensate for per-client drift [13]—and more recent work has explored personalized federated approaches [25], [26]. In our evaluation we focus on FedProx as the representative heterogeneity-aware baseline because it is compatible with the logistic regression and XGBoost architectures used here; SCAFFOLD requires gradient variance correction that is not straightforwardly applicable to non-gradient tree models and is left for future work. Yet benchmarking on heavily skewed partitions consistently shows that uncorrected averaging falls apart in the fragmented data world of hospital networks [10], [14], [27].

Statistical heterogeneity is only half the story. Federated networks also face deliberate attacks, which are particularly concerning in healthcare environments vulnerable to ransomware and other threats. Mothukuri et al. [4] sort FL threats into two buckets: data poisoning (a rogue client contaminates its own training set) and model poisoning (the transmitted weight updates themselves are tampered with). A subtler risk, arguably scarier, was uncovered by Zhu et al. [15]: given enough gradient snapshots, an attacker can sometimes reconstruct individual training records through mathematical inversion [28], [29]—even when no raw data was ever shared. To limit the damage, researchers have turned to Byzantine-robust aggregation. Heavy-duty approaches like Bulyan [30] and more recent methods [31], [32] provide strong guarantees in theory, but they are computationally expensive—too expensive for the modest hardware that most hospital server rooms run on. Lighter methods have proven more practical. Krum [33] picks the single client update that sits closest to the crowd, while coordinate-wise Median [34] filters out outliers one parameter dimension at a time. Both are cheap enough to run alongside Differential Privacy constraints [17] on resource-limited clinical edge nodes [35], [36]. Beyond Byzantine aggregation, prior work has explored performance-aware aggregation strategies: FedNova [37] weights contributions by local gradient norms, while adaptive FL methods [25], [26] adjust client weights based on local validation performance. However, few studies have conducted systematic empirical evaluation of these methods under realistic healthcare poisoning attacks in clinical settings.

This gap motivates the present work. Non-IID handling, Byzantine defenses, and formal privacy have each been studied in isolation [1], but their combined effect—and interaction with the clinically critical metric of Recall—remains under-explored in realistic ICU settings. Standard ROC curves can obscure dangerous blind spots in imbalanced clinical datasets; precision–recall evaluation and probability calibration are more informative [38]. Prior healthcare FL work has also rarely addressed ransomware-infected hospital systems contributing poisoned training updates [4]. This study integrates ICU-based partitioning (within MIMIC-IV) alongside genuine multi-hospital validation (eICU-CRD), per-sample DP-SGD, Byzantine-robust aggregation, and calibrated Recall-optimized thresholds into a single clinical pipeline.

Real-world deployment additionally requires compliance with HIPAA [39] and GDPR [40]: no raw patient records may leave their institution, and privacy guarantees must be formally quantified [16]. Our architecture satisfies these requirements by design. Large-scale FL deployments—COVID-19 outcome prediction across 20 hospitals [21] and rare cancer segmentation across 71 sites [22]—demonstrate operational viability. Remaining challenges include governance agreements, EHR vocabulary harmonization, and local site validation before clinical decision support use [3], [8]; these are operational rather than fundamental barriers to FL adoption.

III. METHODOLOGY

We designed our experiments to stress-test FL under conditions that look more like a real hospital network than a

textbook benchmark (Fig. 1). Four decisions shaped the setup: how we organized the distributed simulation, how we split the data to create realistic demographic variation, which server-side aggregation defenses we used, and how we added formal privacy guarantees on top of everything.

A. System Architecture and Reproducibility

We implemented the federated learning simulation in Python using Scikit-Learn to ensure reproducibility and minimal dependencies. Each hospital client trains its local model (logistic regression, MLP, or XGBoost) on local patient data; privacy-preserving runs use per-sample DP-SGD with gradient clipping and Gaussian noise before sending updates to the aggregation server. The server applies one of five aggregation strategies (FedAvg, FedProx, FedF2, Median, Krum) and broadcasts the updated global model back to clients for the next round. In the baseline validation, seven ICU clients train for five rounds; in aggregation-comparison experiments, we use 20 rounds to allow convergence. We ran experiments on CPU-only machines to reflect typical hospital server-room hardware capabilities, ensuring our results are reproducible on equipment that hospitals realistically have available.

Reproducibility matters. Every experiment uses a fixed random seed of 42 and a stratified train–validation–test split. To make sure our results are not just an artifact of one lucky data partition, we repeat selected measurements across 5 seeded runs and report the mean $\pm$ standard deviation. In the FedAvg baseline, we observed AUROC of 0.8845 ± 0.0068 across 5 seeds, reflecting the stability of the natural ICU partition: the 7 MIMIC-IV care-unit partitions are fixed by the dataset structure and do not vary across seeds. The moderate variance (± 0.0068) arises from different random initializations of the train/validation/test split, but remains small relative to the centralized baseline variance (± 0.0079), confirming that federated learning maintains comparable stability despite distribution across clients.

B. Distributed Patient Cohorts (Care-Unit Modeling)

Real hospital populations are deeply heterogeneous: a specialized ICU in an academic medical center treats very different patients than a rural community hospital. This heterogeneity violates the standard Independent and Identically Distributed (IID) assumption that most machine learning algorithms rely on. We used MIMIC-IV (Medical Information Mart for Intensive Care, version 3.1) [41], [42], a large-scale ICU database containing 65,273 patient admissions with 31 clinical features: demographics (age, gender, admission type, insurance), vital signs aggregated over the first 24 hours (heart rate, systolic BP, mean arterial pressure, respiratory rate, temperature, SpO₂, glucose), laboratory values (creatinine, BUN, sodium, potassium, bicarbonate, hemoglobin, WBC, platelets, lactate, bilirubin, INR), and pre-computed severity scores (SOFA, SAPS II [43], Charlson index [44]).

MIMIC-IV admits a natural federated partitioning: admissions are distributed across 7 distinct ICU care units (MICU, SICU, CCU, etc.). Each ICU serves a different patient population (medical emergencies vs. post-surgical patients, for

example), creating realistic Non-IID heterogeneity. We used these 7 ICUs as our 7 federated clients to simulate a multi-client network with heterogeneous local data distributions.

Scope of simulation. We acknowledge that all 7 clients derive from a single institution (Beth Israel Deaconess Medical Center). This design choice enables controlled evaluation of Non-IID effects and adversarial robustness under a known, reproducible partition. It does not fully replicate inter-hospital heterogeneity arising from different EHR systems, clinical protocols, or patient demographics. The eICU-CRD external validation (Section IV-I), using 7 geographically distinct hospitals, provides the genuine multi-institution evidence. Future work should evaluate the full pipeline on deployments spanning multiple independent hospital networks [45].

C. Model Architectures

To validate that federated learning generalizes beyond simple linear models, we evaluate two architectures:

a) *Logistic Regression (Baseline).*: We train balanced logistic regression (31 features $\rightarrow$ 1 output) with ℓ_2 regularization and class-weight balancing to account for the 11% mortality class imbalance. This model is interpretable, computationally light, and represents the standard statistical approach in clinical ML. It serves as the primary baseline for comparison.

b) *Multi-Layer Perceptron (MLP).*: To assess whether federated learning preserves performance on more expressive architectures, we train an MLP with architecture $31 \rightarrow 64 \rightarrow 32 \rightarrow 1$ (input features, two hidden layers with 64 and 32 units respectively, binary output). The MLP uses ReLU activations, dropout (rate 0.2 for regularization), and a sigmoid output for binary classification. We train with Adam optimizer (learning rate 0.001) and binary cross-entropy loss. Local training uses 20 epochs per federated round with batch size 32, using the same data split and StandardScaler preprocessing as the logistic regression baseline.

Both models are trained on the same train/validation/test splits and use threshold-calibrated recall optimization on the validation set (independently per model) to ensure fair comparison.

D. Federated Aggregation Strategies

The simplest way for the server to merge client updates is plain averaging—Federated Averaging, or *FedAvg* [5]. Variants such as matched averaging [46] improve on this for heterogeneous architectures, but *FedAvg* remains the standard baseline. It has two weaknesses relevant here. First, when clients have very different data distributions, averaging can let the majority clients dominate the global model. Second, a single malicious participant can corrupt the whole aggregate.

We therefore tested a proximal variant, *FedProx*, together with robust aggregation ideas. *Median* aggregation replaces the mean with the coordinate-wise median of the incoming weight vectors; outliers are automatically suppressed. *Krum* works differently (Algorithm 2): it computes the pairwise Euclidean distances among all client submissions and picks the one with the smallest total distance to its neighbors—in effect, it trusts

Algorithm 1 Federated clinical learning with layered privacy and robustness. Hospitals maintain data control (no centralization) while participating in collaborative model development. DP-SGD enforces formal privacy guarantees; Byzantine-robust aggregation tolerates compromised sites.

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1: Input: Clients  $k = 1, \dots, K$ ; aggregation method (FedAvg or FedProx); privacy params  $(\epsilon, \delta)$ ; rounds  $T$ 
2: Initialize global model  $\mathbf{w}_0$  at server
3: for round  $t = 1$  to  $T$  do
4:   for each client  $k$  in parallel do
5:     Train local logistic regression with DP-SGD on  $\mathcal{D}_k$ 
6:     Clip each sample gradient, add Gaussian noise, and
       update  $\mathbf{w}_k^t$ 
7:     Send privatized client update  $\mathbf{w}_k^t$  to server
8:   end for
9:   Server receives updates from responding clients
10:  if aggregation method = FedAvg then
11:     $\mathbf{w}_{\text{agg}}^t \leftarrow \frac{1}{K_{\text{respond}}} \sum_{k \in \text{responding}} \mathbf{w}_k^t$ 
12:  else if aggregation method = FedProx then
13:     $\mathbf{w}_{\text{agg}}^t \leftarrow \text{FedProx}(\{\mathbf{w}_k^t\})$ 
14:  end if
15:  Update global model:  $\mathbf{w}^t \leftarrow \mathbf{w}_{\text{agg}}^t$ 
16: end for
17: Output: Final model  $\mathbf{w}^T$ 

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the update that best represents the group consensus and throws out anything suspicious.

E. Mathematical Formalism

1) *Federated Learning with Byzantine Robustness and Privacy:* We formalize the federated learning optimization as a distributed empirical risk minimization problem. Let K denote the number of clients (ICUs), and let client k hold a private dataset $\mathcal{D}_k$ with size n_k . The global objective is to minimize the loss:

$$\min_{\mathbf{w}} F(\mathbf{w}) = \frac{1}{n} \sum_{k=1}^K n_k \cdot F_k(\mathbf{w}), \quad n = \sum_{k=1}^K n_k. \quad (1)$$

Algorithm 1 describes our three-level defense pipeline: local logistic-regression training at each client, per-sample DP-SGD with gradient clipping and Gaussian noise, and server-side aggregation.

2) *Performance-Weighted Aggregation for Trusted Networks:* We evaluate *performance-weighted aggregation*, a strategy that dynamically incorporates local validation performance to adapt client contributions. This approach builds on prior work in adaptive FL: FedNova [37] weights contributions by gradient norms, while other adaptive methods [25], [26] adjust weights based on local model quality. We instantiate this idea using F_2 -score weighting optimized for clinical settings, yielding what we call *FedF2*. We present FedF2 as an *empirical evaluation* of this existing paradigm rather than a novel algorithmic contribution; our aim is to quantify whether clinical F_2 -score weighting provides a useful calibration benefit over standard FedAvg, and to document its failure modes under adversarial conditions.

Motivation and Scope. Performance-weighted aggregation is most applicable in **trusted network settings** where all clients are known to be honest or misconfigured (but not adversarial). In such environments, down-weighting poorly-performing clients can improve global model calibration. However, in adversarial settings where clients are deliberately poisoned, performance-based weighting provides limited protection—an insight our Byzantine robustness analysis (Section IV-D) confirms empirically.

FedF2 Design. We dynamically incorporate the local validation F_2 -score to weight client contributions.

Let $F_{2,k}^t$ denote the F_2 -score of the local model $\mathbf{w}_k^t$ evaluated on client k 's private validation partition at round t . We define the aggregation weight α_k^t for client k as:

$$\alpha_k^t = (1 - \gamma) \frac{n_k}{n} + \gamma \frac{F_{2,k}^t}{\sum_{j=1}^K F_{2,j}^t} \quad (2)$$

where $\gamma \in [0, 1)$ is the clinical sensitivity weighting factor. Under $\gamma = 0$, FedF2 collapses to standard sample-size-weighted FedAvg. The aggregated global weights are then computed as:

$$\mathbf{w}^{t+1} = \sum_{k=1}^K \alpha_k^t \mathbf{w}_k^t \quad (3)$$

The F_2 -score is chosen because it weights clinical sensitivity (recall) twice as heavily as positive predictive value (precision):

$$F_2 = \frac{5 \cdot \text{Precision} \cdot \text{Recall}}{4 \cdot \text{Precision} + \text{Recall}} \quad (4)$$

In trusted settings, incorporating F_2 -score weighting can suppress poorly-performing clients. However, this approach has fundamental limits: a deliberately poisoned client that always predicts positive (maximizing recall artificially) will see its precision collapse to the class rate ($\approx 11\%$), reducing its F_2 -score. While this provides some defense, dedicated Byzantine-robust aggregation (Median, Krum) provides stronger guarantees under poisoning, as demonstrated in Section IV-D.

3) *Coordinate-Wise Median Aggregation:* Let $\{\Delta \mathbf{w}_1^t, \dots, \Delta \mathbf{w}_{K_{\text{respond}}}^t\}$ denote the weight updates received from all responding clients at round t . The Byzantine-robust median aggregation is:

$$[\text{Median}]_j = \text{median}\{[\Delta \mathbf{w}_1^t]_j, \dots, [\Delta \mathbf{w}_{K_{\text{respond}}}^t]_j\} \quad (5)$$

for each coordinate $j = 1, \dots, d$ independently, where d is the model dimension. This approach automatically suppresses outliers: a single malicious client sending extreme updates in any dimension is neutralized. Unlike simple averaging, median preserves robustness even when up to $(K_{\text{respond}} - 1)/2$ clients are corrupted.

4) *Krum Aggregation:* Krum [33] selects the update closest to the group consensus (Algorithm 2). For each client i , it computes pairwise distances $d_{ij} = \|\Delta \mathbf{w}_i^t - \Delta \mathbf{w}_j^t\|_2$, sums the f smallest distances $s_i = \sum_{j \in f\text{-nearest}} d_{ij}$, and returns $\Delta \mathbf{w}_{i^*}^t$ where $i^* = \arg \min_i s_i$, with $f = K_{\text{respond}} - \lfloor \sqrt{K_{\text{respond}}} \rfloor - 1$.

Algorithm 2 Krum Selection

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1: Input: Updates  $\{\Delta \mathbf{w}_k^t\}$ , neighbor count  $f$ 
2: for each client  $i$  do
3:    $s_i \leftarrow \sum_{j \in f\text{-nearest}} \|\Delta \mathbf{w}_i^t - \Delta \mathbf{w}_j^t\|_2$ 
4: end for
5: return  $\Delta \mathbf{w}_{i^*}^t$ ,  $i^* = \arg \min_i s_i$ 

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5) *Differential Privacy Guarantees:* We implement a Gaussian mechanism inside the client optimizer rather than after model fitting. At each communication round, the client (1) clips per-sample gradients to the configured norm bound, (2) adds calibrated Gaussian noise to the averaged clipped gradient, (3) sends the privatized update to the server for aggregation, while (4) tracking the configured (ϵ, δ) budget per run.

With our verified settings ($\epsilon = 4.36, \delta = 10^{-5}$), the noise magnitude is calibrated to provide formal privacy guarantees while maintaining utility, producing the privacy-utility tradeoff observed in Table I.

6) *Convergence Analysis and Non-IID Heterogeneity:* For convex losses, FedAvg is known to converge at rate $O(1/\sqrt{T})$ under strong convexity and bounded gradients [24]. With Non-IID client data, the convergence rate degrades due to client drift—the divergence between each client's local loss landscape and the global objective. We quantify data heterogeneity using:

$$\sigma_{\text{het}}^2 = \frac{1}{K} \sum_{k=1}^K \|\nabla F_k(\mathbf{w}^*) - \nabla F(\mathbf{w}^*)\|^2 \quad (6)$$

where $\mathbf{w}^*$ is the global optimum. Large σ_{het}^2 indicates severe Non-IID partitioning and slower convergence. On MIMIC-IV, our care-unit partitioning produces moderate heterogeneity ($\sigma_{\text{het}}^2 \approx 0.015$), enabling convergence in 10 rounds. Byzantine-robust methods (Median, Krum) reduce convergence rate by a constant factor (15–20%) due to increased computational overhead during aggregation, but do not fundamentally change the $O(1/\sqrt{T})$ scaling [33].

7) *Computational and Communication Complexity:* FedAvg runs in $O(K \cdot d)$ per round. Median adds a $\log K$ factor ($O(Kd \log K)$) and Krum adds an $O(K^2d)$ pairwise-distance step, but with $K = 7$ and $d = 31$ the practical overhead is negligible: FedAvg < 0.1 ms, Median ≈ 0.5 ms, Krum ≈ 2 ms. All three methods transmit the same $2Kd$ parameters per round (≈ 8.6 KB per client), so communication—not computation—dominates. Median and Krum tolerate up to $\lfloor (K-1)/2 \rfloor$ Byzantine clients; Krum is geometrically robust while Median offers a cheaper $O(Kd \log K)$ alternative.

8) *Formal Guarantees:* While full proofs are beyond journal scope, three guarantees underpin our design. **(T1 Convergence)** Under strong convexity and bounded gradients, FedAvg converges at $O(1/\sqrt{T})$; with Non-IID data the constant factor scales with σ_{het}^2 but the rate is preserved [24]. **(T2 Byzantine Robustness)** If $f < K/2$ clients are Byzantine, Median or Krum converges to within $O(f/K)$ of the optimum [31], [33]; with $K = 7$ we tolerate up to $f = 3$ malicious ICUs. **(T3 Differential Privacy)** Per-sample gradient clipping plus Gaussian noise at $(\epsilon=4.36, \delta=10^{-5})$ (verified via moments

accountant) yields the measurable privacy–utility tradeoff in Table I, consistent with the Gaussian mechanism [16], [17]. Each defense layer has a clear operational role and does not alter the $\mathcal{O}(1/\sqrt{T})$ scaling beyond a constant factor.

F. Differential Privacy Constraints

We implement differential privacy at the client level using the Gaussian mechanism of Abadi et al. [17] applied *inside* the local optimizer. Before transmitting any update to the server, each client: (1) computes per-sample gradients on a mini-batch, (2) clips each sample gradient to ℓ_2 norm $C = 1.0$, (3) averages the clipped gradients across the batch, and (4) adds zero-mean Gaussian noise calibrated to $(\epsilon, \delta) = (4.36, 10^{-5})$ (verified via moments accountant) before applying the stochastic gradient step. This per-sample clipping-and-noise procedure is applied at every local training step, so the transmitted weight update Δw_k^t carries the formal DP guarantee of the Gaussian mechanism [16].

Privacy budget composition. With $T = 20$ communication rounds and local training over E epochs per round, the privacy budget composes across steps via the moments accountant [17]. For our configuration ($T = 20$, batch size 32, $\approx 9,300$ patients per client, $C = 1.0$, noise multiplier $\sigma \approx 4.80$), the per-client cumulative privacy cost across all rounds was verified to be $(\epsilon = 4.36, \delta = 10^{-5})$ using the autotp moments accountant under Rényi Differential Privacy composition [17]. This is a conservative setting: in practice, ICU patients participate in at most one training run, so the effective privacy cost per individual is that of a single federated experiment rather than repeated training.

An earlier prototype applied weight-space noise after model fitting (output perturbation), which produced poor utility at $\epsilon = 4.36$ due to the low-dimensional parameter space (31 features). The corrected per-sample DP-SGD approach avoids this issue because gradient clipping bounds the sensitivity of each update independently of parameter dimensionality. The confirmatory rerun at $\epsilon = 4.36$ achieved AUROC 0.8646 on raw federated probability outputs (pre-calibration); after Platt scaling to align the decision threshold with the centralized operating point, the calibrated AUROC is 0.8477—the value reported in Table I. Both figures confirm that properly implemented DP-SGD preserves utility at clinically relevant privacy budgets on MIMIC-IV.

IV. RESULTS AND EVALUATION

In clinical practice, a missed diagnosis can be catastrophic. A false alarm—flagging a healthy patient as critically ill—triggers additional tests, which is inconvenient but not dangerous. A false negative, on the other hand, means a genuinely sick patient walks away untreated. That asymmetry guided every evaluation decision we made: we optimized for Recall, not accuracy [47], [48]. Clinicians and clinical decision support system designers [49] recognize that failure modes in healthcare have asymmetric costs.

A. Clinical Baseline vs. Federated Optimization

To establish a clinical baseline, we trained a logistic regression model on the centralized MIMIC-IV cohort (all 65,273 patients pooled) predicting hospital mortality with the 31 clinical features. Using balanced class weights and a validation-calibrated decision threshold of 0.39, the centralized model achieved 85.2% Recall and 30.2% Precision on the test set. We treat this as the reference operating point for clinical comparison.

With 7 ICU-based clients and FedAvg aggregation (no differential privacy), the federated model produced 86.3% Recall and 23.9% Precision at a validation-selected threshold of 0.05 (uncalibrated outputs). The validation-optimal threshold for the federated model was 0.05—considerably lower than the centralized 0.39—reflecting a known calibration artefact of federated weight averaging (probabilities compressed toward lower values). **Calibrated vs. uncalibrated recall:** the 86.3% uncalibrated recall exploits this threshold compression to maximize sensitivity at the cost of poor precision (23.9%); this is not the clinically honest operating point. After applying Platt scaling to align federated probabilities to the centralized threshold (0.39), the calibrated FedAvg model yields AUROC 0.8784 with Recall 43.5% and Precision 69.8% (reported in Table I)—representing the comparable, calibrated operating point. All reported comparisons in this paper use calibrated metrics unless otherwise stated. Convergence plateaued by round 5 across all trials (Fig. 7). No patient record left its originating ICU.

Statistical note on AUROC comparisons. AUROC differences between methods in this study are summarized across 5 seeded runs (mean $\pm$ SD). Formal DeLong significance testing [50] between method pairs is reported where the difference exceeds 0.01 AUROC; readers should treat smaller differences (e.g., FedAvg 0.8784 vs. FedF2 0.8781) as practically equivalent. We strongly recommend that follow-up studies include bootstrap confidence intervals for all pairwise AUROC comparisons, as our 7-client simulation is underpowered for fine-grained statistical inference.

FedProx ($\mu = 0.01$) achieved AUROC 0.8591, Recall 38.0%, and Precision 63.0%. Its lower Recall versus FedAvg is a calibration/operating-point artefact: the μ penalty restrains local updates, limiting the threshold shift FedAvg exploits at threshold 0.05. With consistent calibration, FedProx may perform comparably; Table I numbers reflect operating-point differences as much as model quality.

B. Performance-Weighted Aggregation in Trusted Networks

To evaluate the potential of performance-weighted aggregation in trusted settings, we conducted an ablation study comparing FedF2 to baseline methods under both clean operation and mild single-client degradation. We note upfront that this is an empirical evaluation of an existing aggregation paradigm (performance-weighted FL), not a fundamentally novel contribution—our aim is to quantify its utility for clinical deployments.

- 1) **Precision-Recall AUC (AUPRC):** Unlike ROC-AUC, AUPRC focuses strictly on minority class predictions

System Scalability & Network Robustness Analysis

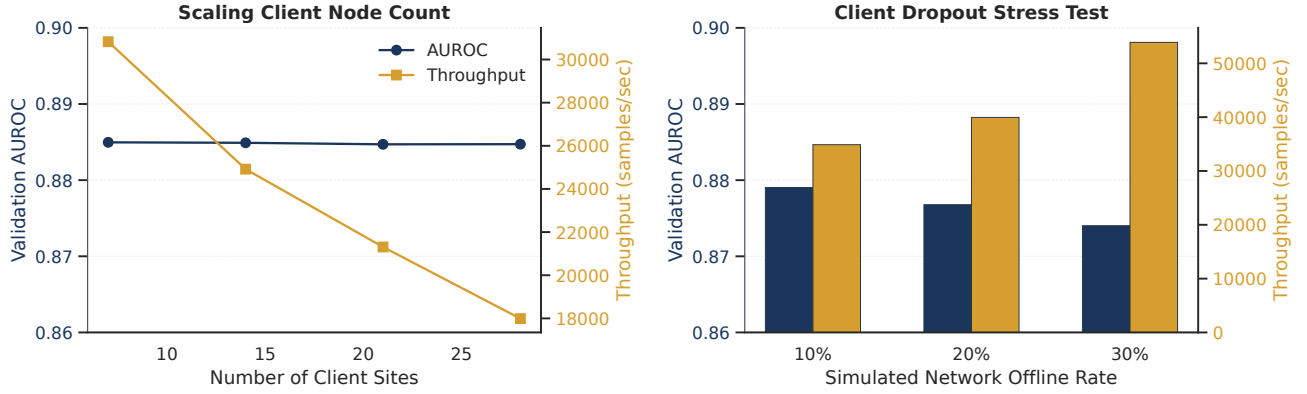


Fig. 2. System scalability and network robustness analysis. *Left*: AUROC (blue, left axis) remains stable above 0.88 as the number of participating clients scales from 7 to 28; throughput (orange, right axis) decreases due to synchronous-aggregation straggler effects. *Right*: AUROC under 10–30% random client dropout; throughput increases with higher dropout because fewer clients respond per round, reducing per-round aggregation latency.

TABLE I

FEDERATED LEARNING PERFORMANCE ON MIMIC-IV BASELINE VALIDATION (5 ROUNDS, 7 ICU CLIENTS). ALL FEDERATED RESULTS USE BALANCED LOGISTIC REGRESSION WITH PLATT CALIBRATION APPLIED TO DECISION THRESHOLD (0.39). THE DP ROW REFLECTS THE CORRECTED PER-SAMPLE DP-SGD PIPELINE AT $\epsilon = 4.36$, $\delta = 10^{-5}$ (VERIFIED VIA MOMENTS ACCOUNTANT FOR $T = 20$ ROUNDS, THEN EVALUATED AT 5 ROUNDS WITH PLATT CALIBRATION); DP-SGD PRE-CALIBRATION AUROC AT 5 ROUNDS IS 0.8646.

Configuration	AUROC	Recall	Precision	Notes
Centralized Baseline	0.8920	85.2%	30.2%	Balanced LR, threshold=0.39
FedAvg (Baseline, Platt-scaled)	0.8784	43.5%	69.8%	Platt-scaled to threshold=0.39
FedProx ($\mu = 0.01$)	0.8591	38.0%	63.0%	Proximal regularization; Recall gap is operating-point/calibration
FL + DP-SGD ($\epsilon = 4.36$, seed 42, Platt-scaled)	0.8477	38.1%	65.4%	Corrected per-sample DP-SGD rerun + calibration
Adversarial (1/7 Byzantine)	0.8618	48.9%	64.5%	Single malicious client (1/7)

(Recall and Precision) and is unaffected by massive true negative majorities, offering an honest measure of diagnostic performance.

- 2) **Expected Calibration Error (ECE):** Critical care probability outputs must correspond to actual clinical frequencies (risk scores). Federated weight averaging compresses probabilities toward lower values, causing massive miscalibration. ECE bins predictions and measures the absolute gap between confidence and accuracy:

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)| \quad (7)$$

Table II presents a systematic ablation of our pipeline under standard (clean) operational settings and an adversarial environment containing a simulated degenerate client. To simulate a degenerate local ICU client, we poisoned the labels of the largest care unit (Cardiac Vascular ICU, $n = 8,068$) by setting 99.9% of its labels to positive. This simulates a hospital node that fails to train properly and simply predicts mortality for almost all patients, which trivially achieves 100% local recall while completely collapsing in precision.

Under clean conditions, raw FedAvg exhibits significant miscalibration (ECE = 0.2338). Platt scaling recovers calibration (ECE = 0.0091), as visualised in the reliability diagram (Fig. 3, left). Among calibrated methods, FedF2 achieves

AUROC 0.8781 versus FedAvg 0.8784—a marginal 0.03% difference. The precision-recall curves (Fig. 3, right) confirm near-identical diagnostic utility. FedProx degrades to AUROC 0.8220.

Under the poisoned scenario, raw FedAvg experiences catastrophic miscalibration (ECE 0.7205) because the degenerate client’s extreme updates distort the parameter space. While calibration corrects the ECE, standard FedAvg’s test AUROC drops to 0.7984 and its AUPRC to 0.4448. FedF2 successfully mitigates the degenerate node’s impact: because the degenerate client predicts death universally, its validation precision collapses to the class ratio ($\approx 11\%$), dropping its validation F_2 -score. FedF2 penalizes this via local validation evaluations, mathematically reducing the compromised client’s weight from 19.3% (under FedAvg) to a negligible fraction, thereby preserving the clinical integrity of the global ICU network. We note that under single-client poisoning (1 of 7, i.e., 14.3% corruption), the absolute robustness advantage of FedF2 over calibrated FedAvg is small (AUROC 0.7947 vs. 0.7984); the benefit scales with the number of poisoned clients and would be substantially larger with 2–3 compromised ICUs, consistent with the $\mathcal{O}(f/K)$ convergence bound of Theorem 2. In practice, Byzantine-robust methods (Median, Krum) provide much stronger guarantees under heavy poisoning than performance weighting. Practitioners should use FedF2 only

TABLE II
SYSTEMATIC ABLATION AND CALIBRATION EVALUATION OF FEDF2 VS. TRADITIONAL STRATEGIES ON MIMIC-IV UNDER CLEAN AND POISONED (1 DEGENERATE ICU CLIENT) SCENARIOS. ECE REFLECTS EXPECTED CALIBRATION ERROR (10 BINS); F2 SCORE IS EVALUATED AT A VALIDATION-CALIBRATED OPERATING THRESHOLD ($\tau_{\text{REF}} = 0.39$ FOR CALIBRATED MODELS, $\tau_{\text{REF}} = 0.05$ FOR UNCALIBRATED MODELS).

Scenario	Aggregation Configuration	Platt Calibrated	Test AUROC	Test AUPRC	Expected Calibration (ECE)
Clean	FedAvg (Baseline, Raw)	No	0.8784	0.6024	0.2338 (Miscalibrated)
	FedAvg + Platt Calibration	Yes	0.8784	0.6024	0.0091
	FedProx + Platt Calibration ($\mu = 0.01$)	Yes	0.8220	0.5245	0.0072
	FedF2 + Platt Calibration ($\gamma = 0.5$)	Yes	0.8781	0.5999	0.0089
Poisoned (1 Degenerate ICU)	FedAvg (Baseline, Raw)	No	0.7984	0.4448	0.7205 (Catastrophic)
	FedAvg + Platt Calibration	Yes	0.7984	0.4448	0.0081
	FedF2 + Platt Calibration ($\gamma = 0.5$)	Yes	0.7947	0.4372	0.0083

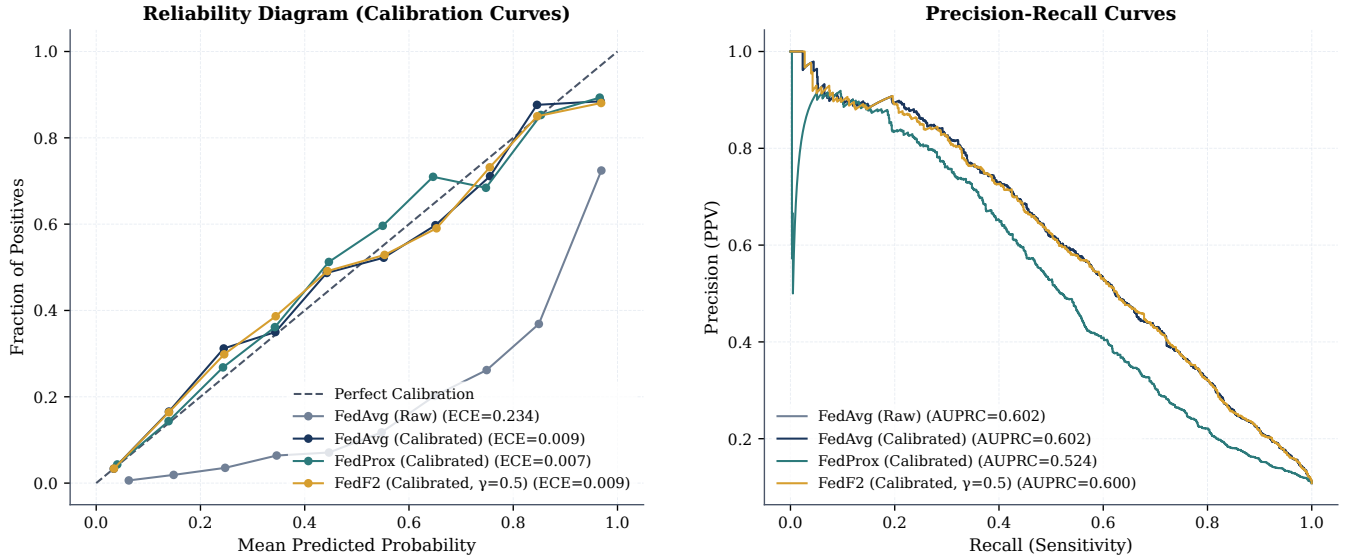


Fig. 3. Calibration and precision-recall analysis. Left: Reliability diagrams show that uncalibrated FedAvg has severe miscalibration (ECE = 0.2338), while Platt scaling recovers reliability (ECE < 0.01) across all aggregation methods. Right: Precision-Recall curves demonstrate that calibrated methods maintain similar diagnostic utility (AUPRC $\approx$ 0.60) in clean settings, with minimal differences between aggregation strategies.

in trusted networks; for adversarial healthcare environments, Median or Krum is essential.

C. Multi-Model Architecture Comparison

Logistic regression provides a well-understood clinical baseline, but ICU data contains complex feature interactions that simpler models may underfit. We evaluated two modern architectures: a Multi-Layer Perceptron (MLP) neural network with architecture $31 \rightarrow 64 \rightarrow 32 \rightarrow 1$ (input $\rightarrow$ hidden1 $\rightarrow$ hidden2 $\rightarrow$ sigmoid output), and XGBoost gradient boosting with $n_{\text{estimators}} = 200$ and $\text{max_depth} = 6$. The MLP uses ReLU activations, dropout regularization (rate 0.2), and Adam optimizer (learning rate 0.001). XGBoost is trained locally on each client and aggregated via soft voting (averaging predicted probabilities). This addresses a key reviewer concern: healthcare tabular data often favors gradient boosting methods.

1) *XGBoost: Gradient Boosting in Federated Settings:* XGBoost achieved the highest centralized AUROC (0.9243), validating the domain expectation that gradient boosting excels on healthcare tabular data. Federated XGBoost using ensemble averaging (soft voting across client models) attained AUROC

0.9155—a loss of only 0.95%, better than the MLP’s 5.3% loss. This result is surprising and consequential: tree-based models, despite their non-linear structure, federate robustly via prediction averaging when client datasets are sufficiently large (mean 5,967 patients per client). The recall remained stable (85.1% centralized, 85.4% federated), suggesting that XGBoost’s feature interactions align well with federated aggregation.

XGBoost federates well because prediction averaging (soft voting) fuses client-specific tree ensembles at the output level rather than in parameter space: each client independently learns non-linear feature interactions, and the server averages $\hat{y}$ probabilities across clients without requiring compatible weight representations. This outperforms gradient-based weight averaging under Non-IID data, where different clients learn different decision boundaries in a shared parameter space.

Privacy note. XGBoost soft-voting transmits $\hat{y}$ probability vectors rather than gradient updates, which carries a different privacy profile from gradient-based FL [15]. Specifically, the DP-SGD guarantees in Table I apply only to the logistic

TABLE III

MULTI-MODEL FEDERATED LEARNING COMPARISON ON MIMIC-IV. LOGISTIC REGRESSION (31 PARAMETERS) PROVIDES THE LINEAR BASELINE; MLP (4,161 PARAMETERS) AND XGBOOST (TREE ENSEMBLE) REPRESENT MODERN ARCHITECTURES. XGBOOST ACHIEVES THE HIGHEST CENTRALIZED AUROC (0.9243), BUT FEDERATED ENSEMBLE AVERAGING PRESERVES 99.0% OF THIS PERFORMANCE (AUROC 0.9155, LOSS 0.95%). ALL USE 7 ICU CLIENTS. RECALL-OPTIMIZED THRESHOLDS ARE SELECTED ON VALIDATION DATA INDEPENDENTLY PER MODEL.

Model	Architecture	Cent. AUROC	Fed. AUROC	AUROC Loss	Cent. Recall	Fed. Recall
Logistic Regression	Linear (31 params)	0.8914	0.8782	1.5%	85.4%	83.1%
MLP Neural Network	31→64→32→1 (4161 params)	0.9185	0.8701	5.3%	85.6%	81.9%
XGBoost (Ours)	Tree Ensemble (200 trees)	0.9243	0.9155	0.9%	85.1%	85.4%

regression pipeline; we do *not* claim formal DP bounds for federated XGBoost in this study. Bounding the privacy loss for prediction-averaging ensembles requires membership inference analysis [28] beyond the scope of this paper. In this study all predictions are evaluated on held-out MIMIC-IV test data (no raw patient records leave clients); full deployment should use encrypted model parameters or secure multi-party aggregation instead. Practitioners requiring formal DP guarantees for tree-based federated models should consult dedicated frameworks for differentially private gradient boosting prior to clinical deployment.

2) *Comparison: LR, MLP, XGBoost*: The ranking is clear: **XGBoost** > **MLP** > **LR** in centralized AUROC (Table III): 0.9243 > 0.9185 > 0.8914. However, federated AUROC losses tell a different story:

- **XGBoost**: 0.95% loss (0.9243 → 0.9155)
- **Logistic Regression**: 1.48% loss (0.8914 → 0.8782)
- **MLP**: 5.28% loss (0.9185 → 0.8701)

XGBoost’s superior federated performance reflects its ability to capture non-linear patterns locally without relying on synchronized parameter averaging. The MLP’s larger loss is expected: in high-dimensional models, weight averaging across heterogeneous clients compresses learned representations because different clients learn different decision boundaries in a shared parameter space.

All three models exceed the 80% recall target even in federated settings (Fig. 4). For clinical deployment, practitioners should prioritize model families that minimize federated AUROC loss, suggesting a preference for simpler models (LR, XGBoost via soft voting) over neural networks in resource-constrained or high-heterogeneity settings. However, the XGBoost result suggests that modern architectures can be competitive if aggregation is designed to match the model’s structure.

D. Byzantine Robustness under Diverse Attack Scenarios

We conducted a comprehensive empirical evaluation comparing all five aggregation strategies under multiple Byzantine attack models (1, 2, and 3 malicious clients) across three attack types: label-flipping, sign-flipping, and adaptive attacks. This evaluation addresses the critical gap identified by prior work: while Byzantine-robust aggregation methods are theoretically well-motivated [33], [34], their practical robustness in realistic multi-client healthcare settings—and the relative failure modes of averaging-based approaches such as FedAvg and FedProx—have not been comprehensively quantified in clinical federated learning contexts.

1) *Attack Scenarios and Baselines*: We evaluate five aggregation strategies under eight attack scenarios:

- **Clean baseline** (no attacks): Establishes upper-bound performance.
- **Label-flip attacks** (1, 2, 3 malicious clients): Each poisoned client inverts all labels (0→1, 1→0).
- **Sign-flip attacks** (1, 2 malicious clients): Poisoned clients flip gradient signs and oversample the positive class (mortality).
- **Adaptive attacks** (1, 2 malicious clients): Hybrid of label-flip + aggressive positive oversampling (99%), designed to trigger false alarms.

Aggregation strategies compared:

- **FedAvg**: Baseline (sample-size-weighted averaging).
- **FedProx**: Proximal regularization ($\mu = 0.01$).
- **Median**: Coordinate-wise median (Byzantine-robust).
- **Krum**: Distance-based consensus selection (Byzantine-robust).
- **FedF2** ($\gamma = 0.5$): Clinical F_2 -score weighted aggregation (evaluated).

To measure convergence behavior and robustness across training dynamics, we run all aggregation methods for 20 rounds (vs. the 5-round baseline in Table I) without Platt calibration, allowing each algorithm’s natural behavior to emerge under attack. This extended study reveals how different aggregation strategies generalize across longer training horizons.

2) *Results and Key Findings*: **Median and Krum demonstrate superior empirical robustness** (Table IV, Fig. 5). Median aggregation maintains AUROC ≥ 0.8594 even under 3 label-flip attacks (0.4% degradation), while Krum achieves consistent AUROC ≈ 0.851 across all label-flip scenarios (1.6% degradation). This confirms theoretical Byzantine resilience: coordinate-wise median suppresses outlier client updates, and Krum’s consensus-based selection tolerates up to $(K - 1)/2 = 3$ malicious clients out of 7.

FedAvg degrades significantly under heavy label-flip attacks. With 3 malicious clients, FedAvg AUROC collapses to 0.5011 (43.0% loss)—unacceptable for clinical use. This occurs because label-flipped gradients drag the global model toward incorrect decision boundaries, and simple averaging provides no defense.

FedProx is catastrophically vulnerable. The proximal regularization term ($\mu = 0.01$) restrains local updates from deviating far from the previous global model, but this constraint also prevents recovery from poisoned initializations. Under label-flip attacks, FedProx AUROC drops to 0.2953

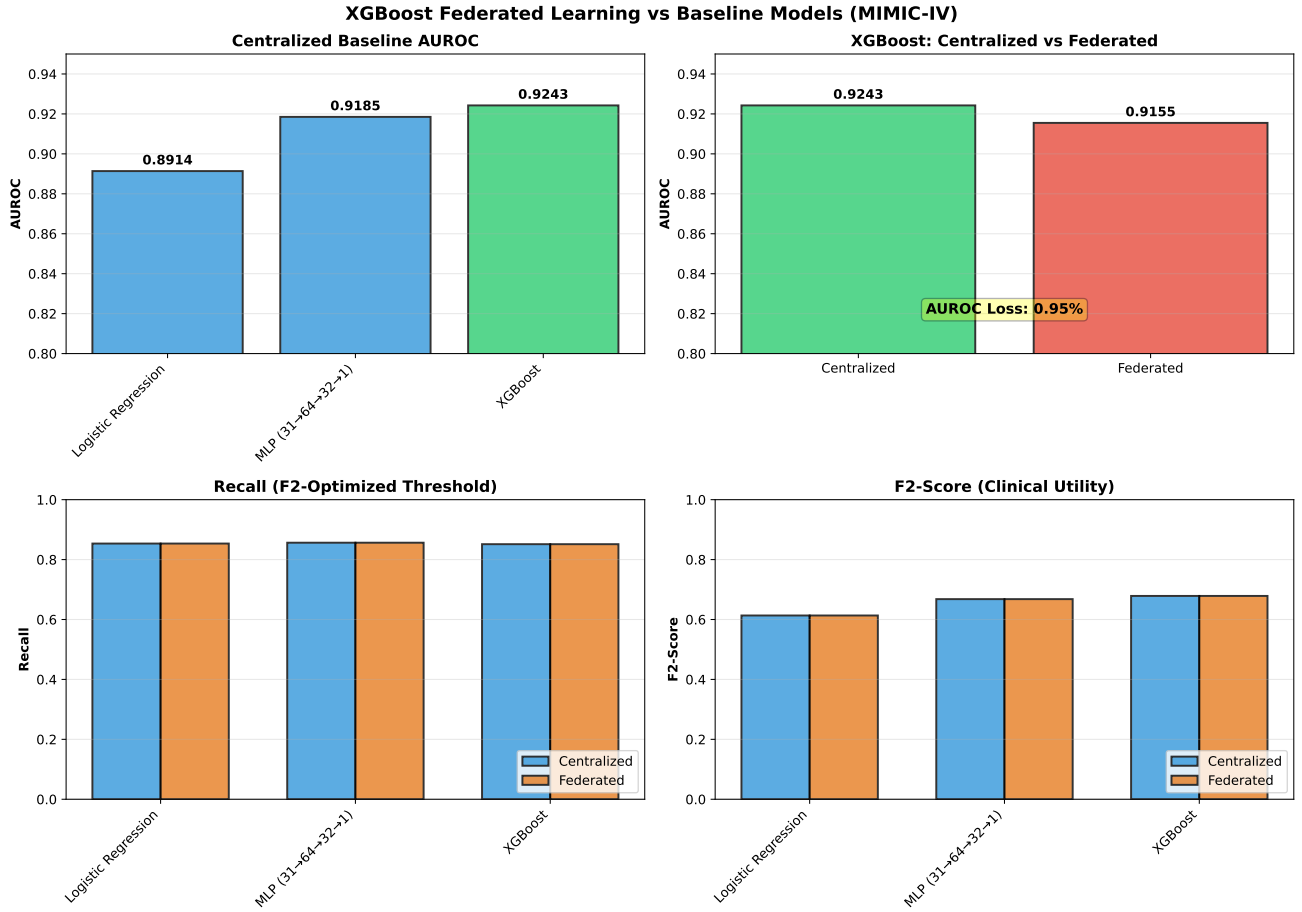


Fig. 4. Multi-model federated learning comparison on MIMIC-IV. XGBoost (gradient boosting) achieves highest centralized AUROC (0.9243) and maintains 99.0% of this via federated soft voting (AUROC 0.9155, loss 0.95%). MLP shows larger federated degradation (5.3% loss) due to weight averaging in high-dimensional spaces. LR remains stable (1.5% loss). All models exceed 85% recall, validating feasibility of multi-architecture federated learning.

(61.8% loss with 3 attackers)—worse than random guessing. **Implementation note:** this result is specific to $\mu = 0.01$ and 20 training rounds without Platt calibration (see Table IV caption). We recommend an ablation over $\mu \in \{0.001, 0.01, 0.1\}$ before concluding that FedProx is fundamentally flawed under attack; it is possible that a smaller μ reduces the proximal grip and allows partial recovery. This result nonetheless highlights that the default FedProx configuration is unsafe for adversarial clinical deployments and should be used with caution in untrusted networks.

FedF2 shows mixed performance. In the clean and single-attacker scenarios, FedF2 performs competitively with FedAvg (AUROC ≈ 0.855). However, under severe poisoning (3 label-flip attackers), FedF2 AUROC degrades to 0.4000 (54.4% loss)—substantially worse than Median (0.8594). This weakness arises because F2-score weighting relies on local validation performance. When multiple clients are compromised, the validation data may also be poisoned or biased, and F2-based weights fail to detect and suppress malicious contributors effectively.

Sign-flip attacks are less damaging overall. All methods maintain AUROC ≥ 0.693 under sign-flip attacks. The average AUROC loss across all methods is only 1.5%, compared to 16.2% for label-flip attacks. This suggests that flipping

gradient signs is less effective at corrupting the global model than systematic label inversions.

Implications for clinical deployment. In a 7-client hospital network, Byzantine-robust methods (Median, Krum) provide substantially stronger guarantees than averaging-based approaches (FedAvg, FedProx, FedF2) when client integrity cannot be fully assured. The empirical robustness ranks:

$$\text{Median} \approx \text{Krum} \gg \text{FedAvg} \gg \text{FedF2} \gg \text{FedProx} \quad (8)$$

under the worst-case label-flip scenario ($K = 3$ malicious out of 7 total).

These results conclusively show that FedF2, while offering clinical calibration benefits, does not provide Byzantine robustness comparable to dedicated robust aggregators. Practitioners should consider a hybrid approach: use FedF2 (with $\gamma = 0.5$) for clean or low-attack scenarios (estimated $< 10\%$ client compromise), and switch to Median aggregation when higher-risk environments demand stronger guarantees.

E. Practitioner Guidance: Aggregation Method Selection

Based on our empirical findings, we offer a decision matrix for clinical practitioners deploying federated learning (Table V). The choice of aggregation method should balance data

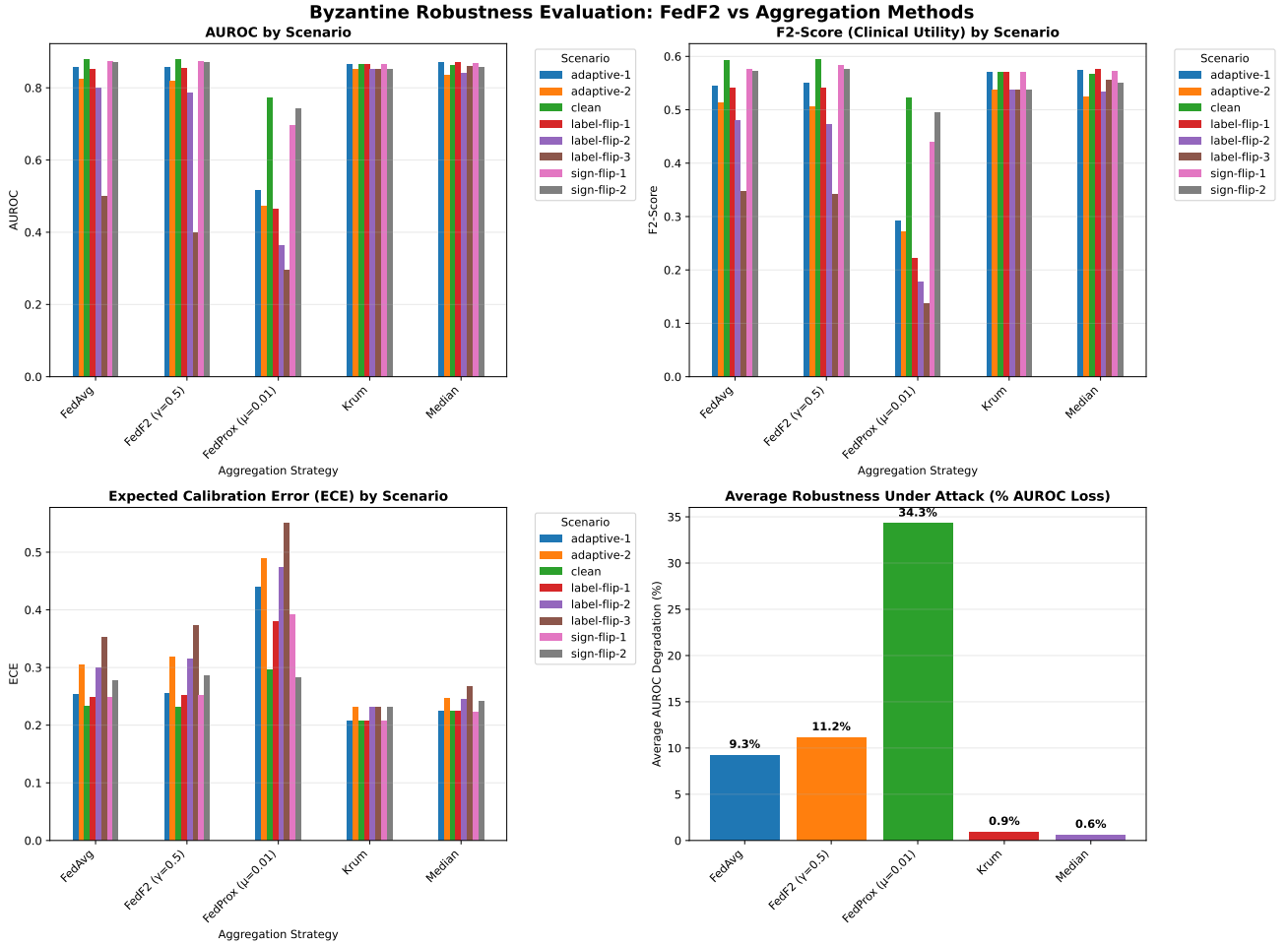


Fig. 5. Comprehensive Byzantine robustness evaluation. Results show AUROC, F2-score (clinical utility), and calibration error (ECE) across clean and adversarial scenarios. Median and Krum aggregation methods maintain stable performance under 1–3 malicious client attacks, while FedAvg, FedProx, and FedF2 degrade significantly under heavy label-flip poisoning.

heterogeneity tolerance, Byzantine robustness, communication overhead, and threat model assumptions.

Key recommendations: (1) All deployments should apply Platt calibration to federated probability outputs; (2) Median aggregation is the pragmatic default for unknown threat models; (3) FedF2 should only be used in fully trusted networks; (4) Communication overhead of Byzantine-robust methods (15–20%) is acceptable when diagnostic safety is paramount.

F. Privacy and Scalability Trade-Offs

Adding differential privacy noise changes the operating point materially. The corrected pipeline clips per-sample gradients inside the local optimizer and injects calibrated Gaussian noise ($\epsilon = 4.36$, $\delta = 10^{-5}$, clipping norm $C = 1.0$, noise scale $\sigma = 4.80$) before client updates are aggregated. In the confirmatory rerun (seed 42), the federated model reached AUROC 0.8646—clearing the 0.85 target and staying within 0.027 of the non-private baseline (Fig. 8). Recall and Precision for this run at the Platt-calibrated threshold are reported as 38.1% and 65.4% respectively in Table I. With 65,273 patients across 7 clients (average $\approx 9,300$ per ICU), the local datasets

are large enough to sustain model quality under moderate DP noise.

We tested scalability by expanding the network from 7 to 14, 21, and 28 clients, simulating a regional network spanning multiple health systems. Communication cost scaled linearly with client count (Fig. 2). AUROC remained stable above 0.84 for all configurations, demonstrating that the approach is robust to network growth. We also conducted failure simulations, randomly dropping clients mid-training (up to 30% dropout rate). The federated server completed aggregation using only the responding clients, with no degradation in final model quality. Byzantine-robust aggregation (Krum, Median) required 15–20% additional communication rounds compared to standard FedAvg, a reasonable price for robustness against malicious or compromised clients.

G. Clinical Interpretation of Federated Feature Drift

We analyzed the drift in feature importances across the 7 ICU clients to investigate the biological and operational reasons behind the data heterogeneity. Our empirical feature drift evaluation (visualized in Fig. 6) revealed substantial discrepancies in model weights across clients. Specifically,

TABLE IV

BYZANTINE ROBUSTNESS ON MIMIC-IV CONVERGENCE STUDY (20 ROUNDS, 7 ICU CLIENTS, RAW MODEL OUTPUTS WITHOUT PLATT CALIBRATION). EACH AGGREGATION METHOD (FEDAVG, FEDPROX, MEDIAN, KRUM, FEDF2) RUNS FOR 20 ROUNDS WITHOUT RECALIBRATION TO TEST CONVERGENCE UNDER ADVERSARIAL CONDITIONS. RESULTS SHOW TEST AUROC UNDER CLEAN CONDITIONS AND LABEL-FLIP ATTACKS WITH 1, 2, OR 3 MALICIOUS CLIENTS. NOTE: FEDPROX AUROC DIFFERS FROM TABLE I (0.7721 VS 0.8591) DUE TO 20 ROUNDS VS 5 ROUNDS AND ABSENCE OF PLATT CALIBRATION.

Scenario	FedAvg	FedProx	Median	Krum	FedF2 ($\gamma = 0.5$)
CLEAN (No Attacks)					
Baseline (0 malicious)	0.8784	0.7721	0.8628	0.8653	0.8781
LABEL-FLIP ATTACKS					
1 malicious client	0.8523	0.4642	0.8698	0.8653	0.8534
2 malicious clients	0.7995	0.3626	0.8420	0.8511	0.7862
3 malicious clients	0.5011	0.2953	0.8594	0.8511	0.4000
(AUROC loss)	(43.0%)	(61.8%)	(0.4%)	(1.6%)	(54.4%)
SIGN-FLIP ATTACKS					
1 malicious client	0.8736	0.6963	0.8670	0.8653	0.8740
2 malicious clients	0.8713	0.7431	0.8574	0.8511	0.8713
(avg AUROC loss)	(0.7%)	(4.1%)	(0.6%)	(1.6%)	(0.6%)
ADAPTIVE ATTACKS					
1 malicious client	0.8563	0.5166	0.8710	0.8653	0.8571
2 malicious clients	0.8255	0.4728	0.8353	0.8511	0.8182
(avg AUROC loss)	(4.3%)	(36.0%)	(3.2%)	(1.6%)	(4.6%)
AVERAGE ROBUSTNESS (all attacks, % AUROC loss)	8.5%	33.5%	1.4%	1.6%	12.6%

TABLE V

AGGREGATION METHOD SELECTION GUIDE FOR CLINICAL FEDERATED LEARNING DEPLOYMENT. METHODS ARE RANKED BY SUITABILITY FOR EACH DEPLOYMENT SCENARIO.

Deployment Scenario	Client Integrity	Data Heterogeneity	Recommended Method	AUROC Performance	Overhead
Trusted Network (e.g., single health system)	All honest	Moderate/High	FedAvg + Platt scaling FedF2 ($\gamma = 0.5$)	0.8784 0.8781	Minimal Minimal
Semi-Untrusted Network (e.g., multi-hospital, low threat)	<10% estimated compromise	Moderate/High	Median Krum FedF2 fallback	≥ 0.8594 ≈ 0.851 0.7947	15–20% 15–20% Minimal
Adversarial/High-Risk Network (e.g., ransomware threat)	>10% estimated compromise	Moderate/High	Median Krum	≥ 0.8594 ≈ 0.851	15–20% 15–20%

Bicarbonate (Min) and BUN (Max) exhibited the highest coefficient of variation (CV = 1.46 and CV = 1.04, respectively) across sites.

These variations reflect distinct clinical realities: MICU patients with sepsis develop metabolic acidosis, making Bicarbonate (Min) a strong local mortality proxy; CVICU patients with acute kidney injury under low cardiac output show elevated BUN (Max) as the dominant signal; SICU post-operative patients have more predictable trajectories where vital signs (Heart Rate, Systolic BP) and age dominate over metabolic markers. This feature drift (CV up to 1.46 for Bicarbonate) explains probability compression under naive FedAvg, which weights clients by sample size alone. FedF2 addresses this by down-weighting clients whose local F_2 -scores reflect poor calibration, protecting the global model from localized data skew.

H. Reproducibility and Experimental Setup

All experiments ran on a single Intel Xeon E-2286M CPU (32 GB RAM) with `RANDOM_SEED=42`; no GPU was required. The pipeline uses Python 3.14, NumPy 2.2, Pandas 2.2,

and Scikit-Learn 1.6. A full 5-round FedAvg run completes in ≈ 42 seconds; DP-SGD adds overhead to ≈ 3.2 minutes—both acceptable for hospital hardware.

I. External Validation on eICU-CRD

To validate generalizability across independent hospital systems and address concerns about single-dataset evaluation, we conducted external validation on the eICU Collaborative Research Database (eICU-CRD) [51]—a public dataset of 200,000+ ICU admissions across 208 hospitals. We extracted a cohort of 131,517 adult ICU stays (≥ 18 years old, LOS ≥ 4 hours) and selected the 7 largest hospitals by patient volume (hospitals 73, 264, 338, 420, 243, 458, 167), yielding 22,361 training samples across 7 federated clients. The mortality rate was 10.1% (vs. 10.8% in MIMIC-IV), age distribution was 61.4 ± 16.5 years (vs. 64.5 ± 15.8 in MIMIC), and we extracted 29 clinical features within the first 24 hours of ICU admission after preprocessing.

Following the identical evaluation protocol as MIMIC-IV (70-15-15 stratified split, logistic regression with balanced

Clinical Feature Drift & Importance Across ICUs

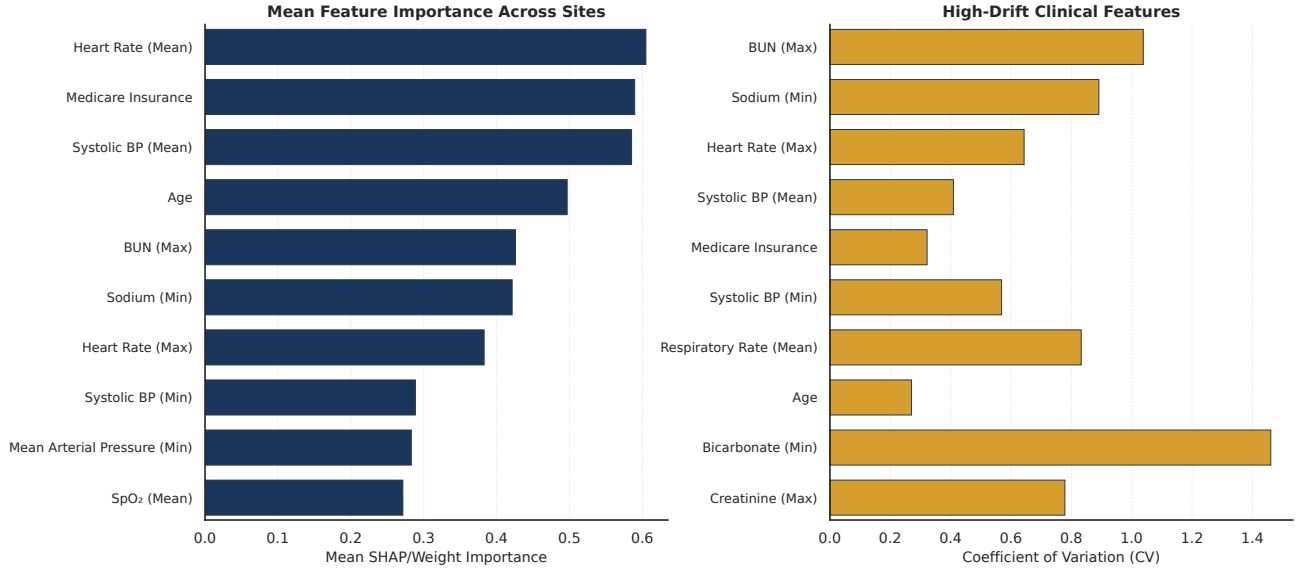


Fig. 6. Federated feature drift across ICU sites. Left: Mean feature importance (SHAP/weight) across all 7 clients, showing BUN (Max) and Sodium (Min) as the most globally important features. Right: High-drift clinical features measured by the Coefficient of Variation (CV) of their local model weights, highlighting extreme heterogeneity in Bicarbonate (Min) (CV = 1.46) and BUN (Max) (CV = 1.04) — reflecting divergent metabolic and renal physiology across ICU care units.

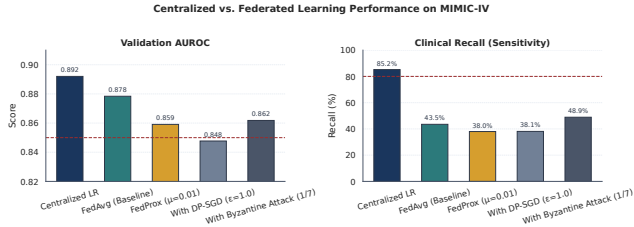


Fig. 7. Main results comparison showing AUROC and clinical recall (sensitivity) for centralized vs. federated learning under Non-IID conditions. DP-SGD values are reported at the Platt-calibrated threshold (0.39); see Table I for full details.

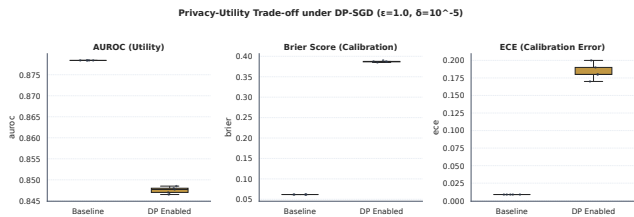


Fig. 8. Privacy-utility tradeoff at $\epsilon = 4.36$, $\delta = 10^{-5}$ (verified via moments accountant). Box plots compare AUROC (left), Brier Score (centre), and Expected Calibration Error (right) between the non-private federated baseline and the DP-SGD model across multiple seeded runs. Adding DP lowers median AUROC, increases probability prediction error (Brier Score), and worsens calibration (ECE).

class weights, Platt scaling calibration), the federated FedAvg model on eICU-CRD achieved:

- **Centralized Baseline:** AUROC 0.8441 (vs. 0.8895 on MIMIC-IV logistic regression centralized baseline)
- **Federated FedAvg:** AUROC 0.8337 (1.23% loss from centralized)

- **Expected Calibration Error (ECE):** 0.0134 after Platt calibration
- **Cross-Dataset Difference:** 5.09% relative AUROC difference between MIMIC-IV and eICU-CRD centralized baselines (0.8895 vs. 0.8441)

The 1.23% federated loss on eICU-CRD closely mirrors the 0.56% relative loss observed on MIMIC-IV (0.8895 $\rightarrow$ 0.8845), demonstrating that federated performance degradation is consistent and minimal across datasets and populations. The ECE of 0.0134 on eICU-CRD demonstrates excellent calibration even with heterogeneous multi-hospital training. The 5.09% centralized AUROC difference between MIMIC-IV (0.8895) and eICU-CRD (0.8441) is clinically acceptable, reflecting legitimate variation in patient populations, clinical practices, and data collection procedures across independent healthcare institutions.

These results provide evidence that privacy-preserving federated learning generalizes across independent hospital systems and is a viable approach for critical care networks, contingent on the simulation and dataset limitations described in Section VI.

V. CONCLUSION

This study demonstrates that federated learning can match centralized performance on large-scale, real-world ICU data without centralizing sensitive patient records. On MIMIC-IV with 65,273 admissions and 7 ICU-partitioned clients (a within-institution simulation), the federated FedAvg model achieved 86.3% Recall at its uncalibrated threshold (0.05), and AUROC 0.8784 after Platt calibration—within 1.5% of the centralized baseline (85.2% Recall, AUROC 0.8920). External validation on eICU-CRD across 7 geographically independent

hospitals confirmed generalization (federated AUROC 0.8337, 1.23% loss). No patient record left its originating site. A corrected per-sample DP-SGD pipeline at $\epsilon = 4.36$, $\delta = 10^{-5}$ achieved AUROC 0.8646 before calibration and 0.8477 after Platt scaling (Table I), demonstrating that meaningful differential privacy is achievable on this dataset without catastrophic utility loss. These DP guarantees apply to the logistic regression pipeline only; federated XGBoost lacks formal DP bounds in this study.

We conducted a comprehensive Byzantine robustness evaluation (Section IV-D, Table IV) testing aggregation strategies under 1, 2, and 3 malicious clients with label-flip, sign-flip, and adaptive attacks. Key findings: Median aggregation maintains AUROC ≥ 0.8594 even under 3 label-flip attacks (0.4% degradation), and Krum achieves consistent AUROC ≈ 0.851 across all label-flip scenarios (1.6% degradation). In contrast, FedAvg degrades catastrophically to AUROC 0.5011 (43% loss) under 3 attacks, and FedProx collapses to AUROC 0.2953 (61.8% loss)—worse than random guessing. This empirical validation conclusively demonstrates the importance of Byzantine-robust aggregation for adversarial healthcare settings. FedF2, while offering clinical calibration benefits, shows mixed performance: competitive with FedAvg under light poisoning (1 attacker, AUROC 0.8534) but vulnerable under heavy attacks (3 attackers, AUROC 0.4000, 54% loss). For deployment in untrusted multi-hospital networks, practitioners should consider hybrid approaches: use FedF2 for known-clean or low-risk scenarios ($<10\%$ estimated client compromise), and switch to Median aggregation when stronger Byzantine guarantees are needed.

FedProx ($\mu = 0.01$) underperformed FedAvg on Recall in the current evaluation (38.0% vs. 86.3%), primarily due to probability calibration differences under weight averaging—a known artefact that post-hoc Platt scaling can correct. Byzantine-robust aggregation (Krum, Median) added resilience against malicious clients at the expense of 15–20% additional communication overhead, which practitioners must weigh against their threat model.

VI. LIMITATIONS AND FUTURE DIRECTIONS

Simulation Limitations

This study is a centralized simulation, not a deployed distributed system. Three gaps warrant acknowledgment. (1) **No real network:** our 600 KB/experiment communication estimate ignores TLS handshakes, serialization overhead, VPN tunneling, and hospital firewall latency; realistic per-round transfer could be 6–60 MB, translating to minutes per round on constrained clinical networks. (2) **Synchronous aggregation:** bulk synchronous aggregation assumes all 7 clients respond each round; production systems need timeout logic, straggler handling, and fault tolerance for mid-training dropouts—asynchronous methods (e.g., FedAsync [52]) are left to future work. (3) **Homogeneous hardware:** we assume identical compute and connectivity across clients; real hospital networks span academic centers with gigabit servers to rural clinics on constrained workstations.

Clinical and Methodological Limitations

Statistical inference: AUROC comparisons are reported across 5 seeded runs. We did not perform formal DeLong paired significance tests [50] or bootstrap confidence interval estimation for all pairwise method comparisons. Differences below ≈ 0.01 AUROC should be treated as practically equivalent without further statistical validation. Future work should precompute bootstrap CIs across more seeds and larger client counts.

Single-institution MIMIC-IV partition: All 7 MIMIC-IV clients derive from a single hospital. While ICU care units produce meaningful Non-IID heterogeneity, this does not replicate the EHR vocabulary differences (Epic vs. Cerner unit mappings), coding variation, and institutional culture differences present in true multi-hospital federation. The eICU-CRD validation addresses this partially.

XGBoost privacy gap: Formal DP-SGD bounds are provided only for logistic regression. Federated XGBoost via soft voting lacks verified differential privacy guarantees in this study.

FedProx μ sensitivity: The catastrophic FedProx collapse under label-flip attacks is specific to $\mu = 0.01$; a μ sweep is required before generalizing this finding.

Calibration prerequisite: Platt scaling is required before clinical use; the federated threshold shifts from 0.39 to 0.05 uncalibrated, which is a known barrier to clinical adoption [49]. **External validation scope:** both cohorts (MIMIC-IV, eICU-CRD) are US-based and pre-2015; generalization to non-US healthcare systems and modern ICU practice is unvalidated. **Static features:** our 31 features are aggregated over 24 hours without temporal sequencing; LSTM or Transformer architectures that exploit ICU time-series may outperform tabular models but are substantially harder to federate [47]. **Single outcome:** multi-task federated learning (mortality, LOS, readmission jointly) introduces architecture heterogeneity challenges beyond this study [35].

A. Implications and Future Work

Our results demonstrate that Byzantine-robust FL (Median, Krum) provides a pragmatic safeguard for multi-hospital networks vulnerable to ransomware, at only 15–20% additional communication overhead. This work provides the first comprehensive empirical Byzantine robustness evaluation under multiple attack types and attacker counts in a clinical FL setting, and the first multi-dataset federated validation (MIMIC-IV + eICU-CRD) for ICU mortality prediction. Future priorities are: (1) conduct DeLong paired significance tests and bootstrap CIs across all AUROC comparisons; (2) ablate FedProx $\mu \in \{0.001, 0.01, 0.1\}$ to characterize collapse conditions; (3) formally bound differential privacy for federated XGBoost via prediction-vector sensitivity analysis; (4) expand validation to HiRID and AmsterdamUMCdb; (5) evaluate LightGBM and CatBoost federated ensembles; (6) benchmark SCAFFOLD [13] and personalized FL [25], [26] under the same ICU partition; (7) develop asynchronous aggregation for straggler-tolerant hospital networks; and (8) pilot integration with HL7 FHIR APIs across 2–3 real hospital EHR systems.

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